Student number_____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

The Second Semester of 2020—2021 Year

Applied Calculus III Final Examination

(Closed book Time 120 minutes)

Question number	I	II	III	IV	V	Total score
Score						
Marker						

I. Completions (5 items, 3 points each)

Score	

1. The angle θ between two planes $2x-y+z-7=0$ and $x+y+2z-11=0$ is ____.

2. Suppose that equation $z=f(xz,yz)$ defines z implicitly as a function about x and y, and f is continuously differentiable, then $\frac{\partial z}{\partial x}=$_____.

3. Evaluate the integral $\int_0^2 dx\int_x^2 e^{-y^2}dy=$_____.

4. Evaluate the integral $\int_C zdx+xdy+ydz=$_____, where $C: x=t^2, y=t^3, z=t^2$, t is from 0 to 1.

5. The interval of convergence of the series $\sum_{n=0}^{+\infty}\frac{(-1)^n x^n}{n+1}$ is_____.

II. Choices (5 items, 3 points each)

Score	

6. Existence of the partial derivatives f_x and f_y at some point is the ( ) condition for the differentiability of function $f(x,y)$ at such point. ( )

A. sufficient. B. necessary.

C. sufficient and necessary. D. neither sufficient nor necessary.

7. Find which of the following statements is FALSE ( )

A. Two different lines parallel to a third line are parallel.

B. Two different lines perpendicular to a plane are parallel.

C. Two different planes parallel to a line are parallel.

D. Two different planes perpendicular to a line are parallel.

8. Let $f(x)$ be continuous. Then $\int_0^{\frac{\pi}{2}} d\theta \int_0^{\cos\theta} f(r\cos\theta, r\sin\theta) r dr$ is equal to ( ).

A. $\int_0^{\frac{1}{2}} dy \int_0^1 f(x,y)dx$

B. $\int_0^1 dx \int_0^{\sqrt{x-x^2}} f(x,y)dy$

C. $\int_0^1 dx \int_0^{\sqrt{2x-x^2}} f(x,y)dy$

D. $\int_0^{\frac{1}{2}} dy \int_{\frac{1}{2}-\sqrt{\frac{1}{4}-y^2}}^{\frac{1}{2}+\sqrt{\frac{1}{4}-y^2}} f(x,y)dx.$

9. Suppose $L: y = x^2 (0 \le x \le \sqrt{2})$. then $\int_L x ds =$( ).

A. 2 B. 0 C. $\frac{13}{6}$ D. $\frac{5}{6}$

10. If the power series $\sum_{n=1}^{+\infty} a_n x^n$ is convergent at the point x_0, then when $|x| < |x_0|$ the power series $\sum_{n=1}^{+\infty} a_n x^n$ is ( ).

A. absolutely convergent B. divergent

C. conditionally convergent D. none of above

III. Calculations and Proofs (6 items, 8 points each)

Score	

11. Determine the convergence or divergence of the series $\sum_{n=2}^{+\infty} \frac{(-1)^n}{\sqrt{n^2 - n}}$. If it is convergent, please determine whether it is conditionally or absolutely convergent, and give the reason.

12. Evaluate the limit $\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{xy+4}-2}$.

13. Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 2$.

14. Evaluate the line integral $\int_L \frac{xdy - ydx}{4x^2 + y^2}$, where L is a circle with the center $(1, 0)$ and radius $R(R > 1)$, and the counter-clockwise direction.

15. Evaluate the triple integral $\iiint_V (x^2+y^2+z^2)dxdydz$, where V is the unit ball $x^2+y^2+z^2 \le 1$.

16. Let $z=(x^2+y^2)\ e^{-arctan\frac{y}{x}}$. Find dz and $\frac{\partial^2 z}{\partial x \partial y}$.

IV. Analytical calculations（2 items, 9 points each）

Score	

17. Let the fluid velocity $\vec{v}(x,y,z)=\{xy,yz,zx\}$. Find the outward flux of the fluid across the surface of the solid region Ω bounded by planes $z=1, x=0, y=0$ and the cone $z=\sqrt{x^2+y^2}$ (Suppose that the fluid density is 1) .

18. Find the convergence interval and the sum of the power series $\sum_{n=2}^{+\infty} n(n-1)x^n$.

V. Discussion Problem（4 points）

Score	

19. We've learned a lot of concepts in《Applied Calculus III》.These concepts are the basis for some important properties and theorems. What concepts have impressed you most? Please list at least 3 concepts, choose one of them and give their definitions, and briefly explain why you are impressed.